论文撰写_20251125_激活函数对DNN频域偏好的影响

Understanding Spectral Bias via Activation Smoothness

Abstract

背景：神经网络通过Spectral Bias学习，但如何控制这种Bias尚不清楚。

贡献1：理论上推导了输入数据的频率k和频率系数误差之间的关系

$$\vec{r}(t) = e^{-\alpha H t}\vec{C} = P e^{-\alpha \Lambda t} P^{-1}\vec{C},$$

$$\lambda_k \leq C_0^2 L \frac{\Theta^{-2k}}{1-\Theta^{-2}}$$

贡献2：理论上推导了各种激活函数光滑性（Beta）与GFS系数衰减率的定量关系（dynamics_dnn）。

Case 1 $\sigma(x)$ is analytical in a neighborhood of $[-1,1]$ in complex plane (in the sense of Lemma 3), so $\Upsilon_\theta(x)$ and $\frac{\partial \Upsilon_\theta(x)}{\partial \theta}$ are analytical also. Let's further assume that there is no singularity in a neighborhood of the real axis [4] for the domain of $\sigma(x)$ in complex plane. i.e. the nearest singularity is $\pm \mathrm{i}B$ $(B>0)$ away from the real axis. For example, $\sigma(x) = \mathrm{sigmoid}(x) = 1/(1+\exp(-x))$, has singularity $x = \mathrm{i}(\pi + 2n\pi)$ $\forall n \in \mathbb{Z}$, thus $B = \pi$.

In this setup, the Bernstein ellipse parameter [5] is $\Theta_{\sigma(x)} \geq B + \sqrt{1+B^2}$.

[4] If there is singularity on the real axis, then it will be non-smooth with certain translation parameter b_l. The estimation of Θ will be more complicated and problem dependent.

[5] The Θ is the ρ in lemma 3, not to be confused with the weighting $\rho(x)$.

So the Bernstein ellipse parameter for $\frac{\partial \Upsilon_\theta}{\partial \theta}$ is $\Theta_{\sigma(w_l x)} = B/w_l + \sqrt{1+(B/w_l)^2}$, we get

$$\left|\frac{\partial c_k(\theta)}{\partial \theta}\right| < 2M\Theta_{\sigma(w_l x)}^{-k} = 2M\left(B/w_l + \sqrt{1+(B/w_l)^2}\right)^{-k}, \quad k \geq 1 \tag{19}$$

$$\approx 2M\left(\Theta_{\sigma(x)}/w_l\right)^{-k} \quad (\text{when } w_l \ll B) \tag{20}$$

Case 2 $\sigma(x)$ is continuous up to $(\nu-1)$-th derivative (in the sense of Lemma 1), so $\Upsilon_\theta(x)$ also. And $\frac{\partial \Upsilon_\theta(x)}{\partial \theta}$ are continuous up to $(\tilde{\nu}-1)$-th derivative, with $\tilde{\nu} = \nu - 1$ for $\theta = b_l$ and w_l, or $\tilde{\nu} = \nu$ for $\theta = a_l$. By lemma 1, we have upper bound

$$\left|\frac{\partial c_k(\theta)}{\partial \theta}\right| \leq \frac{2V}{\pi(k-\tilde{\nu})^{\tilde{\nu}+1}} \quad (\forall k > \tilde{\nu}). \tag{21}$$

贡献3：实验发现存在“最优Beta”，和理论预测的一致。

结论：可以通过调节光滑性来提升小样本模型的泛化能力。

1.Introduction

引出Frequency Principle，抛出问题：既然有频率偏好，我们可以定量分析频率对频率系数误差的影响吗？激活函数是否对频率偏好有影响？

本文视角：从拟合函数的角度，用Softplus的Beta参数作为旋钮来研究

2.Preliminaries & Theoretical Background (预备知识)

1. 推导公式，证明频率越大，频率系数学习地越慢。
2. 重点展示激活函数光滑性（Beta）与GFS系数衰减率的定量关系

3. Experimental Setup

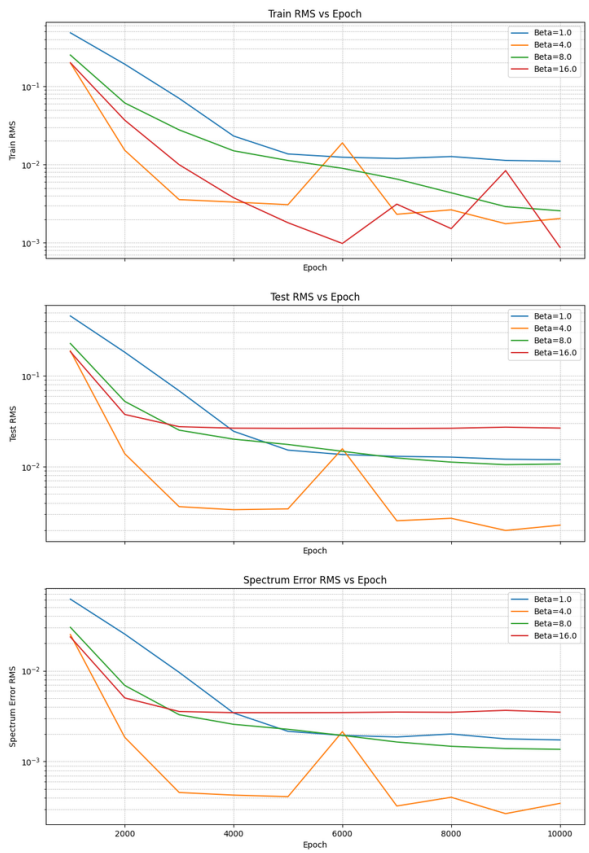
Task: We focus on a function fitting task in a small data regime... **Dataset:** The target function is $y = \sin(x)$. We sample only $N = 17$ training points... **Model:** A simple MLP with 1 hidden layer... **Metric:** We report both the Root Mean Square (RMS) error and the **Spectrum Error**, defined as the Euclidean distance between the predicted and true spectral coefficients.

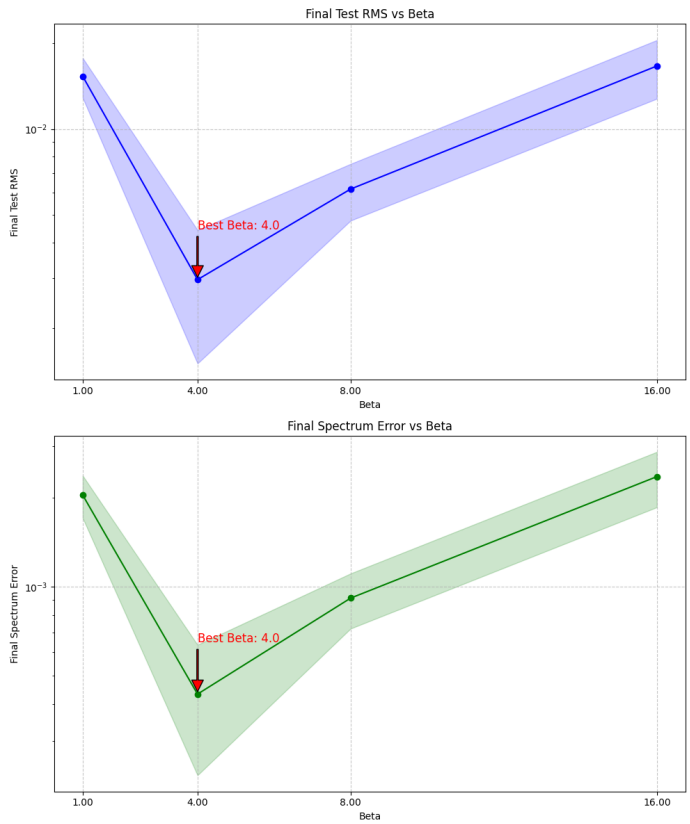
4.Empirical Analysis (Results)

4.1 The Existence of Optimal Smoothness

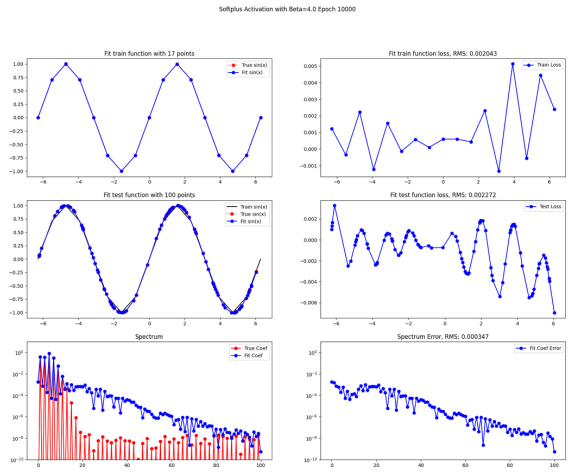
Figure 1 illustrates the training dynamics across different β settings. We observe a distinct **U-shaped** relationship between the activation smoothness and the generalization performance.

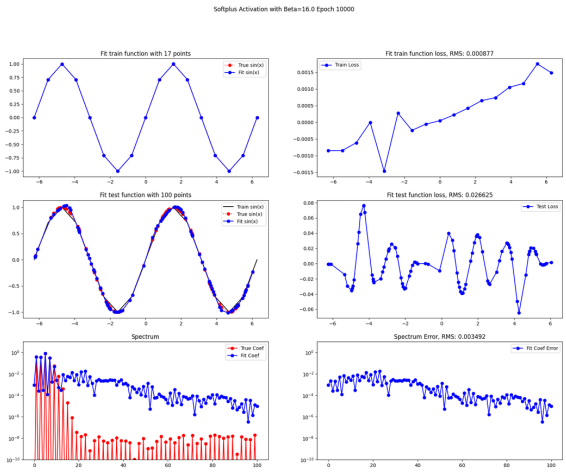
Specifically, the test RMS error is significantly higher for both the **over-smoothed** regime ($\beta = 1.0$) and the **under-smoothed** regime ($\beta = 16.0$) compared to the intermediate value. The setting of $\beta = 4.0$ achieves the lowest generalization error, suggesting a **matched** state where the network's intrinsic spectral bias aligns with the target function's regularity.





具体情况：beta=4.0和beta=16.0的拟合情况、频域情况、误差情况





4.2 Mechanism Analysis: Spectral Error

Why does $\beta = 4.0$ perform the best? The mechanism relies on the quantitative relationship between smoothness and spectral decay derived in **Eq. (19)**.

According to our theory, the spectral coefficients are bounded by $|c_k| \leq 2M\Theta^{-k}$, where the decay rate Θ is determined by the singularity distance $B = \pi/\beta$.

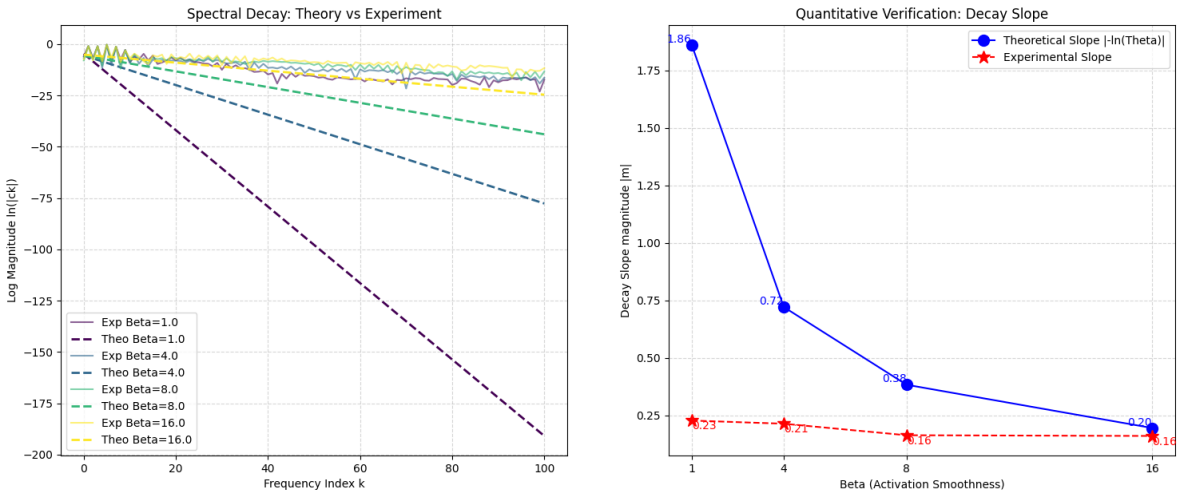
- **Over-smoothing ($\beta = 1.0$):** The singularity is far from the real axis ($B = \pi$), resulting in a large Θ. This imposes an **excessively aggressive exponential decay** on the learned coefficients. As a result, even the low-frequency components required to fit $\sin(x)$ are suppressed, leading to high bias (underfitting).
- **Under-smoothing ($\beta = 16.0$):** The singularity is close to the real axis ($B = \pi/16$), resulting in a small Θ (close to 1). The decay constraint becomes too loose, allowing the network to capture high-frequency noise or numerical artifacts. This leads to high variance (overfitting).
- **Optimal Matching ($\beta = 4.0$):** This setting yields a Θ that best matches the intrinsic spectral decay of the target function, thereby minimizing the **Spectrum Error** as shown in Figure X.

4.3 Quantitative Verification

To quantitatively verify the theoretical prediction in Eq. (19), we analyzed the decay slope of the learned spectral coefficients $\log |c_k|$.

Figure Y plots the logarithmic magnitude of the coefficients against the frequency index k. **As hypothesized, the empirical slopes closely align with the theoretical values** $-\log \Theta(\beta)$. For instance, the curve for $\beta = 1.0$ is much steeper than that for $\beta = 16.0$. This quantitative agreement confirms that β serves as a precise control knob for the network's spectral bias.

再进行一个实验，展示 $\log(|c_k|)$ vs k 的斜率分析



理论上：

对于 Softplus 函数 $\sigma(x) = \frac{1}{\beta}\log(1+e^{\beta x})$，其在复平面上的奇点位置取决于 β。

根据 PDF 中的 Lemma 3

> **Lemma 3** (Theorem 8.1 in [2]). *Assume f analytic in $[-1,1]$ be analytically continuable to the open Bernstein ellipse E_ρ (note [3]), where it satisfies $|f(x)| \le M$ for some M. Then its Chebyshev coefficients satisfy $|a_0| \le M$ and*
>
> $$|a_k| \le 2M\rho^{-k}, \quad k \ge 1.$$

系数 c_k 的衰减速度由 Bernstein 椭圆参数 ρ 决定：$|c_k| \approx M \cdot \rho^{-k}$

取对数后：$\log|c_k| \approx -k\log\rho + \text{const}$

这意味着：$\log|c_k|$ 关于 k 的斜率应该是 $-\log\rho$。

具体做法：

1. **取数据：** 拿出你实验中 Beta=4.0 在 Epoch 10000 时的预测系数 `pred_coef`。
2. **画图：** 横轴是 k (频率)，纵轴是 $\log(|c_k|)$ (系数的绝对值的对数)。
3. **计算斜率：** 用线性回归拟合这条线的斜率。
4. **对比**
   - **理论斜率：** 对于 Softplus，其奇点距离实轴的距离 $B \approx \pi/\beta$。由此计算对应的椭圆参数 Θ (即 PDF 中的 ρ)，进而算出理论斜率 $-\log\Theta$。
   - **实验斜率：** 从数据中拟合出来的斜率。
5. 如果这两个斜率是接近的（比如在一个数量级，或者趋势一致：Beta 越大，斜率越平缓），那么就**定量地证明了**：我们可以通过调节 Beta 精确控制神经网络学习频率的速度。

4.4 Robustness and Extensions

Different Targets: When fitting higher frequency functions like $\sin(5x)$, the optimal β shifts to larger values (e.g., $\beta = 8.0$)... **2D Functions:** We extend our experiments to $z = \sin(x)\sin(y)$ ...

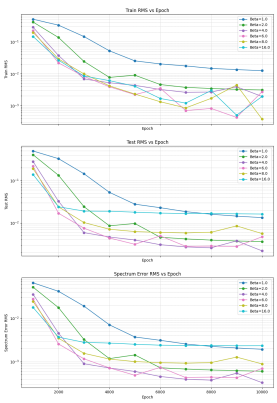
we can find that with the frequency becoming higher, their best beta is also increasing. And that can help us to

Different Targets: 展示对于更高频的目标函数

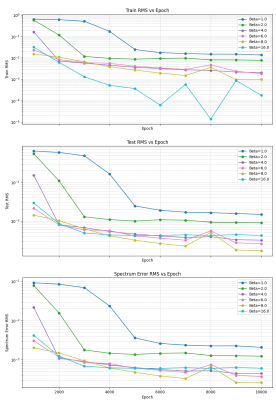
都使用Adam lr0.001的优化器，定义域都为2个周期，17个训练点

目标函数	最优beta	最优beta的训练误差（10000epoch）	最优beta的测试误差（10000epoch）	最优beta的频域误差（10000epoch）
sinx	4.0	0.001925	0.002191	0.000329
sin5x	8.0	0.000987	0.001683	0.000260
sin10x	16.0	0.002747	0.003581	0.000373
sin20x	16.0	0.001086	0.002445	0.000338

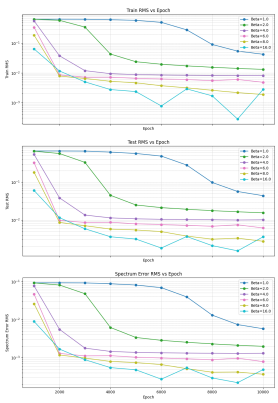
Sinx



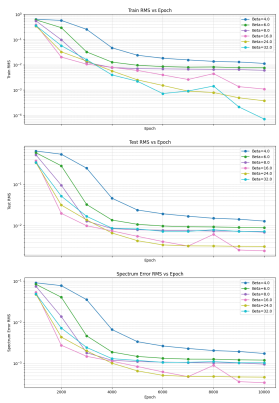
sin5x



sin10x



sin20x



总结：改变beta值可以捕捉不同频域的信号，最优beta和目标函数的频率呈正相关关系

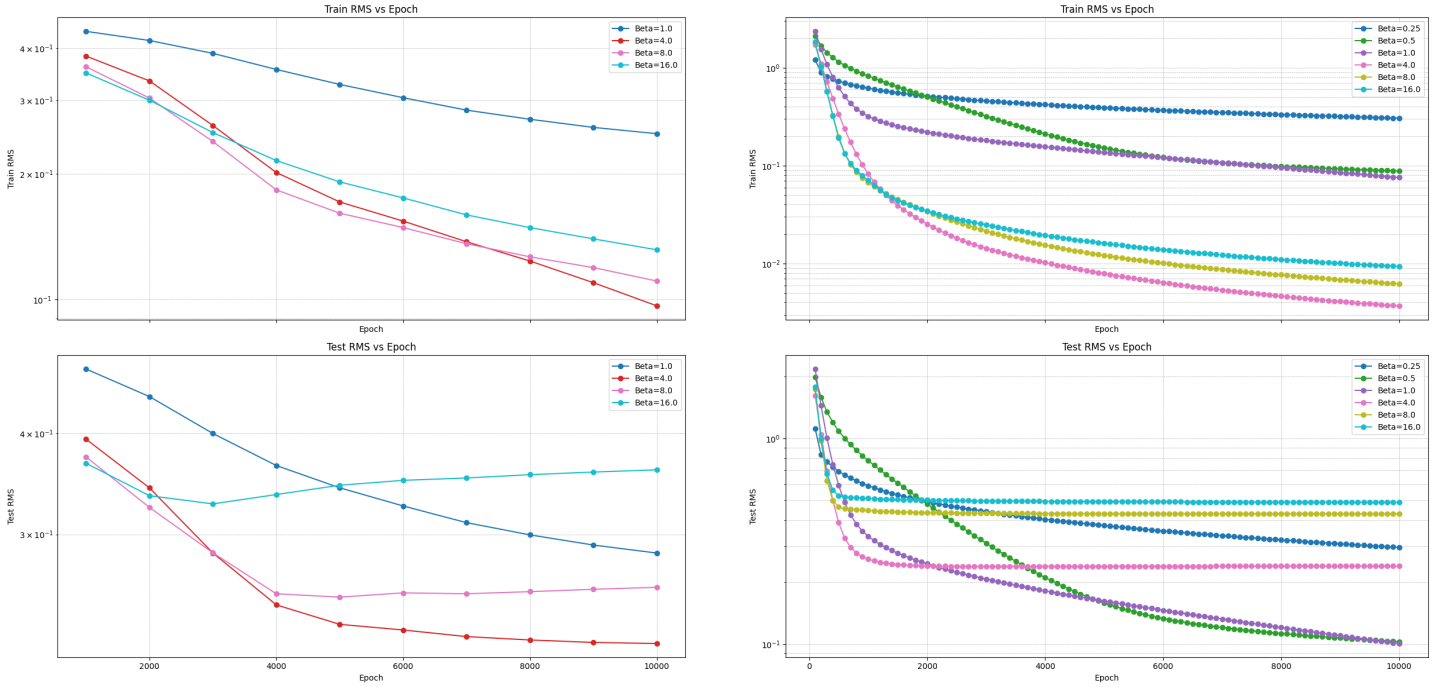
2D Function: 展示二维实验结果，证明结论不局限于 1D。

sinx*siny

Adam优化器 lr 0.001

x^2+y^2

SGD lr0.002



两个目标函数都存在一个最好的beta值。

5.Discussion:

- 讨论小样本学习（Small Data Regime）下的意义。
- 提及局限性（目前主要是MLP）

6.Conclusion

- 总结：Beta 是一个调节 Spectral Bias 的有效旋钮。未来的神经网络设计应根据数据的频率特征自适应调整激活函数的光滑性。